Lecture - 8

❖ Integrated Development Environment:

An IDE (Integrated Development Environment) is a software application that combines all the basic tools a programmer needs to write and test code into one single program.

- **Key Components of an IDE:**
  - ✓ **Code Editor:**
    A place to write and edit your text code with features like color-coding (syntax highlighting) and auto-complete.
  - ✓ **Compiler or Interpreter:** A tool that translates your human-readable code into language the computer can run.
  - ✓ **Debugger:** A tool that helps find and fix errors or bugs in the code.
  - ✓ **Terminal/Console:** An integrated window to see the output or run commands directly.